

$$\mathbf{p} = \hat{n} \tan(\omega/2)$$

$$\omega = 2 \arccos(q_0)$$

$$\mathbf{p} = \hat{n} \tan(\omega/2)$$

$\mathbf{q}$

$$R(\phi_1, \Phi, \phi_2)$$

$$(\phi_1, \Phi, \phi_2)$$

Φ

ω

$SO(3)$

$$\mathbf{q} = q_0 + q_1 \mathbf{i} + q_2 \mathbf{j} + q_3 \mathbf{k}$$

S^3

$$R(\phi_1, \Phi, \phi_2)$$

R

$$|\mathbf{q}| = 1$$

$$\hat{n} = (n_1, n_2, n_3)$$

$$\mathbf{q} = q_0 + q_1 \mathbf{i} + q_2 \mathbf{j} + q_3 \mathbf{k}_{SO(3)}$$

$$|\mathbf{q}| = 1$$

$$\hat{n} = (n_1, n_2, n_3)$$

S^3